

Lab 8: Linux Basics Part 8

Network Scanning & Enumeration with Nmap

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Introduction: This lab dives into practical network reconnaissance and enumeration using Nmap, a key tool in the industry used for network discovery and security auditing. It progresses from host discovery to advanced service and vulnerability enumeration. Even practicing active reconnaissance, running multiple scan types, and fingerprinting services as well as operating systems. This lab also helped me utilize the Nmap Scripting Engine (NSE) to automate vulnerability checks and learn to interpret their results. Throughout the exercises I focus on collecting accurate, reproducible evidence while managing scan timing and stealth on my two designated targets, Linux (Metasploitable) and Windows (Metasploitable3).

Exercise 1:

```

(quintin@kali)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:4b:a8:c7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.103/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 6526sec preferred_lft 6526sec
    inet6 fe80::a00:27ff:fe4b:a8c7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(quintin@kali)-[~]
$ cd ~/cybersec-labs

(quintin@kali)-[~/cybersec-labs]
$ mkdir nmap-lab

(quintin@kali)-[~/cybersec-labs]
$ cd nmap-lab

(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ 

```

Press Enter to run verbose scan...

```

Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-28 19:48 EDT
Initiating SYN Stealth Scan at 19:48
Scanning localhost (127.0.0.1) [1000 ports]
Completed SYN Stealth Scan at 19:48, 0.04s elapsed (1000 total ports)
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000050s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

```

```

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
      Raw packets sent: 1000 (44.000KB) | Rcvd: 2000 (84.000KB)

```

Notice the difference?

- Verbose mode shows scan progress in real-time
- You see 'Discovered open port ...' as each port is found
- More statistics at the end

This is why flags are useful - they modify Nmap's behavior!

Practice complete! Now you're ready for
the exercises that follow.

```

Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 19.22% done; ETC: 19:50 (0:00:04 remaining)
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 27.84% done; ETC: 19:50 (0:00:03 remaining)
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 37.65% done; ETC: 19:50 (0:00:02 remaining)
Nmap scan report for pfSense.home.arp (192.168.1.1)
Host is up (0.0026s latency).
MAC Address: 08:00:27:B7:66:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.100
Host is up (0.0015s latency).
MAC Address: 08:00:27:52:28:68 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.103
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.00 seconds

(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ cat host_discovery.txt
# Nmap 7.95 scan initiated Tue Oct 28 19:50:09 2025 as: /usr/lib/nmap/nmap --privileged -sn -oN host_discovery.txt 192.168.1.0/24
Nmap scan report for pfSense.home.arp (192.168.1.1)
Host is up (0.0026s latency).
MAC Address: 08:00:27:B7:66:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.100
Host is up (0.0015s latency).
MAC Address: 08:00:27:52:28:68 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.103
Host is up.
# Nmap done at Tue Oct 28 19:50:11 2025 -- 256 IP addresses (3 hosts up) scanned in 2.00 seconds

(quintin@kali)-[~/cybersec-labs/nmap-lab]
$

```

```

(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ sudo nmap -PR 192.168.1.0/24
[sudo] password for quintin:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-28 19:56 EDT
Nmap scan report for pfSense.home.arp (192.168.1.1)
Host is up (0.0017s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:B7:66:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.1.100
Host is up (0.00089s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry

```

```

(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ sudo nmap -sn --min-rate 1000 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-28 20:01 EDT
Nmap scan report for pfSense.home.arpa (192.168.1.1)
Host is up (0.0019s latency).
MAC Address: 08:00:27:B7:66:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.100
Host is up (0.0015s latency).
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.103
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 0.78 seconds

(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ nmap -sn --min-rate 1000 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-28 20:02 EDT
Nmap scan report for pfSense.home.arpa (192.168.1.1)
Host is up (0.0016s latency).
MAC Address: 08:00:27:B7:66:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.100
Host is up (0.0011s latency).
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.103
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 0.76 seconds

```

Checkpoint Questions:

1. Yes -sn skips port scan and only identifies hosts
2. -sn is just host discovery and regular port scan is host discovery + port scanning
3. Yes

Exercise 2:

Session Actions Edit View Help

MAC Address: 08:00:27:B7:66:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.100
Host is up (0.0018s latency).
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.103
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 1.98 seconds

≡ Detailed Discovery ≡

Trying multiple discovery methods ...
Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-28 20:08 EDT
Nmap scan report for pfSense.home.arpa (192.168.1.1)
Host is up (0.0015s latency).
MAC Address: 08:00:27:B7:66:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.100
Host is up (0.0018s latency).
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.103
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.04 seconds

Results saved to discovery_detailed.txt

≡ How to identify your targets ≡

Metasploitable (Linux): Look for host with many open ports(21,22,23,80 ...)
Metasploitable3 (Windows): Look for Windows ports (135,139,445,3389)

Next step: Pick two IPs to use as targets and update targets.txt

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
```

```
$ nmap $TARGET1_IP
```

```
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 13:00 EDT
```

```
Nmap scan report for 192.168.1.100
```

```
Host is up (0.0014s latency).
```

```
Not shown: 977 closed tcp ports (reset)
```

PORT	STATE	SERVICE
21/tcp	open	ftp
22/tcp	open	ssh
23/tcp	open	telnet
25/tcp	open	smtp
53/tcp	open	domain
80/tcp	open	http
111/tcp	open	rpcbind
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
512/tcp	open	exec
513/tcp	open	login
514/tcp	open	shell
1099/tcp	open	rmiregistry
1524/tcp	open	ingreslock
2049/tcp	open	nfs
2121/tcp	open	ccproxy-ftp
3306/tcp	open	mysql
5432/tcp	open	postgresql
5900/tcp	open	vnc
6000/tcp	open	X11
6667/tcp	open	irc
8009/tcp	open	ajp13
8180/tcp	open	unknown

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ cat metasploitable_basic.txt
# Nmap 7.95 scan initiated Wed Oct 29 13:02:26 2025 as: /usr/lib/nmap/nmap --privileged -oN metasploitable_basic.txt 192.168.1.100
Nmap scan report for 192.168.1.100
Host is up (0.0034s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]  
$ grep "open" metasploitable_basic.txt | wc -l  
23
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]  
$ grep "open" metasploitable_basic.txt  
21/tcp    open    ftp  
22/tcp    open    ssh  
23/tcp    open    telnet  
25/tcp    open    smtp  
53/tcp    open    domain  
80/tcp    open    http  
111/tcp   open    rpcbind  
139/tcp   open    netbios-ssn  
445/tcp   open    microsoft-ds  
512/tcp   open    exec  
513/tcp   open    login  
514/tcp   open    shell  
1099/tcp  open    rmiregistry  
1524/tcp  open    ingreslock  
2049/tcp  open    nfs  
2121/tcp  open    ccproxy-ftp  
3306/tcp  open    mysql  
5432/tcp  open    postgresql  
5900/tcp  open    vnc  
6000/tcp  open    X11  
6667/tcp  open    irc  
8009/tcp  open    ajp13  
8180/tcp  open    unknown
```



```
GNU nano 8.6 scan_analysis.txt
Port Scan Analysis - Exercise 2
-----
Target 1 IP: [your_IP]
Quick Scan Results:
- Number of open ports:23
- Interesting services found:
* Port 21 (FTP):
* Port 22 (SSH):
* Port 80 (HTTP):
Questions to consider:
- Why is FTP on port 21 a concern? FTP isn't a secure transfer protocol because it transmits in plaintext,
making it vulnerable to packet sniffing
- Is telnet (port 23) secure? Why/why not? No telnet isn't secure on port 23 because it is obsolete
and has been replaced by ssh on port 22. Telnet also transmits in plaintext making it vulnerable to not only packet sniffing
but telnet sessions can also be hijacked by attackers
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ nmap -p- $TARGET1_IP -oN metasploitable_allports.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 13:19 EDT
Nmap scan report for 192.168.1.100
Host is up (0.00093s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ nmap -p- -T4 --min-rate 1000 $TARGET1_IP
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 13:36 EDT
Nmap scan report for 192.168.1.100
Host is up (0.00083s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ echo "=== Quick Scan Results ==="
grep "open" metasploitable_basic.txt | wc -l
echo "open ports found"
echo ""
echo "=== Full Scan Results ==="
grep "open" metasploitable_allports.txt | wc -l
echo "open ports found"
=== Quick Scan Results ===
23
open ports found

=== Full Scan Results ===
30
open ports found
```

```

nmap -p 80,443,8080,8443 $TARGET1_IP
== Scanning common services ==
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 14:03 EDT
Nmap scan report for 192.168.1.100
Host is up (0.0019s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
80/tcp    open  http
443/tcp   closed https
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
== Scanning first 1000 ports ==
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 14:04 EDT
Nmap scan report for 192.168.1.100
Host is up (0.0021s latency).
Not shown: 988 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec

```

Checkpoint Questions:

1. Open = port is accepting connections, filtered = firewall is most likely blocking probe
2. There are other important processes that run on ports past the first 1000 ports
3. -p flag specifies the specific port to scan/listen/connect to

Exercise 3:


```
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh
80/tcp  open  http
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Nmap done: 1 IP address (1 host up) scanned in 0.37 seconds

=== With Version Detection (-sV flag) ===

Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-29 20:33 EDT
Nmap scan report for 192.168.1.100
Host is up (0.013s latency).

```
PORT    STATE SERVICE VERSION
21/tcp  open  ftp      vsftpd 2.3.4
22/tcp  open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
80/tcp  open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 7.38 seconds
Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-29 20:33 EDT
Nmap scan report for 192.168.1.100
Host is up (0.0011s latency).

Not shown: 977 closed tcp ports (reset)

```
PORT    STATE SERVICE VERSION
21/tcp  open  ftp      vsftpd 2.3.4
22/tcp  open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp  open  telnet   Linux telnetd
25/tcp  open  smtp     Postfix smtpd
```

(quintin@kali)-[~/cybersec-labs/nmap-lab]

```
$ nmap -sV --version-intensity 0 -p 21,22,80 $TARGET1_IP
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 20:44 EDT
Nmap scan report for 192.168.1.100
Host is up (0.0018s latency).
```

```
PORT    STATE SERVICE VERSION
21/tcp  open  ftp      vsftpd 2.3.4
22/tcp  open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
80/tcp  open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 6.53 seconds

(quintin@kali)-[~/cybersec-labs/nmap-lab]

```
$ nmap -sV --version-intensity 5 -p 21,22,80 $TARGET1_IP
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-30 17:12 EDT
Nmap scan report for 192.168.1.100
Host is up (0.0016s latency).
```

```
PORT    STATE SERVICE VERSION
21/tcp  open  ftp      vsftpd 2.3.4
22/tcp  open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
80/tcp  open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 6.51 seconds

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
```

```
$ nmap -sV --version-all -p 21 $TARGET1_IP
```

```
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-30 17:14 EDT
```

```
Nmap scan report for 192.168.1.100
```

```
Host is up (0.00097s latency).
```

```
PORT      STATE SERVICE VERSION
```

```
21/tcp    open  ftp      vsftpd 2.3.4
```

```
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

```
Service Info: OS: Unix
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

```
Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
```

```
$ TARGET2_IP="192.168.1.104"
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
```

```
$ nmap -sV -p 135,139,445,3389 $TARGET2_IP
```

```
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-30 17:26 EDT
```

```
Nmap scan report for 192.168.1.104
```

```
Host is up (0.0017s latency).
```

```
PORT      STATE SERVICE      VERSION
```

```
135/tcp    open  msrpc        Microsoft Windows RPC
```

```
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
```

```
445/tcp    open  microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
```

```
3389/tcp    open  tcpwrapped
```

```
MAC Address: 08:00:27:D7:CC:D8 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

```
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

```
Nmap done: 1 IP address (1 host up) scanned in 7.60 seconds
```

```
SF:1\x20505\x20HTTP\x20Version\x20Not\x20Supported\r\nDate:\x20Fri,\x2031\x20Oct\x202025\x2002:51:54\x20GMT\r\nConnection:\x20close\r\nContent-Length:\x200\r\n\r\n");  
MAC Address: 08:00:27:D7:CC:D8 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows
```

```
Host script results:
```

```
| smb2-time:  
|   date: 2025-10-31T02:55:15  
|_  start_date: 2025-10-30T21:23:38  
| smb2-security-mode:  
|   2.1:0:  
|_  Message signing enabled but not required  
|_  clock-skew: mean: 1h02m12s, deviation: 2h38m41s, median: 2m21s  
| smb-os-discovery:  
|   OS: Windows Server 2008 R2 Standard 7601 Service Pack 1 (Windows Server 2008 R2 Standard 6.1)  
|   OS CPE: cpe:/o:microsoft:windows_server_2008::sp1  
|   Computer name: vagrant-2008R2  
|   NetBIOS computer name: VAGRANT-2008R2\x00  
|   Workgroup: WORKGROUP\x00  
|_  System time: 2025-10-30T19:55:16-07:00  
|_ nbstat: NetBIOS name: VAGRANT-2008R2, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:d7:cc:d8 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
| smb-security-mode:  
|   account_used: <blank>  
|   authentication_level: user  
|   challenge_response: supported  
|_  message_signing: disabled (dangerous, but default)
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

```
Nmap done: 1 IP address (1 host up) scanned in 222.31 seconds
```

```
└─$ nmap -sV -sC -p- $TARGET2_IP -oN metasploitable3_detailed.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-30 22:50 EDT
Nmap scan report for 192.168.1.104
Host is up (0.00069s latency).
Not shown: 65502 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
21/tcp    open  ftp              Microsoft ftpd
| ftp-syst:
|_  SYST: Windows_NT
80/tcp    open  http             Microsoft IIS httpd 7.5
| http-methods:
|_  Potentially risky methods: TRACE
|_http-server-header: Microsoft-IIS/7.5
|_http-title: Site doesn't have a title (text/html).
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds     Windows Server 2008 R2 Standard 7601 Service Pack 1 microsoft-ds
1617/tcp  open  java-rmi         Java RMI
| rmi-dumpregistry:
|_  jmxrmi
|     javax.management.remote.rmi.RMIServerImpl_Stub
|     @192.168.1.104:49177
|     extends
|         java.rmi.server.RemoteStub
|         extends
|             java.rmi.server.RemoteObject
3306/tcp  open  mysql            MySQL 5.5.20-log
| mysql-info:
|_  Protocol: 10
|_  Version: 5.5.20-log
|_  Thread ID: 6
```



```
(quintin@kali)-[~/cybersec-labs/nmap-lab]  
$ ./banner_grab.sh
```

Banner Grabbing Exercise

What is a banner?

- A message sent by a service when you connect
- Often reveals software name and version
- Can be grabbed manually with netcat or automatically with Nmap

— Manual FTP Banner (Target 1) —

Command: nc -w 2 192.168.1.100 21

What to expect: FTP server will announce itself

220 (vsFTPd 2.3.4)

^C

— Manual SSH Banner (Target 1) —

Command: nc -w 2 192.168.1.100 22

What to expect: SSH version string

SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1

— Manual HTTP Banner (Target 1) —

Command: Send HTTP request to get server header

What to expect: Web server type and version

HTTP/1.1 200 OK

Date: Wed, 29 Oct 2025 05:12:40 GMT

Server: Apache/2.2.8 (Ubuntu) DAV/2

Now let's compare with Nmap's version detection

— Nmap Service Detection (Target 1) —

Command: nmap -sV --version-intensity 5 -p 21,22,80 192.168.1.100

Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-30 23:16 EDT

Nmap scan report for 192.168.1.100

Host is up (0.0018s latency).

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

21/tcp	open	ftp	vsftpd 2.3.4
--------	------	-----	--------------

22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
--------	------	-----	--

80/tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
--------	------	------	-------------------------------------

MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 6.51 seconds

— Nmap Service Detection (Target 2 - Windows) —

Command: nmap -sV -p 445,3389 192.168.1.104

Starting Nmap 7.95 (<https://nmap.org>) at 2025-10-30 23:16 EDT

Nmap scan report for 192.168.1.104

Host is up (0.0011s latency).

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

445/tcp	open	microsoft-ds	Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
---------	------	--------------	--

3389/tcp	open	ms-wbt-server?	
----------	------	----------------	--

GNU nano 8.6

service_analysis.txt

Service Version Analysis

TARGET 1 (Linux - Metasploitable):

Port 21 (FTP):

Service name: FTP vsFTPD 2.3.4

Version detected: vsFTPD 2.3.4

Banner received: 220 vsFTPD 2.3.4

Potential vulnerabilities: Backdoor command

Next steps: Use :) login to gain unauthorized root access

Port 22 (SSH):

Service name: OpenSSH

Version detected: 4.7

Banner received: SSH-2.0 OpenSSH_4.7p1 Debian-8ubuntu

Security notes: very old, several known vulnerabilities

Next steps: use known vulnerabilities to exploit outdated OpenSSH service

Port 80 (HTTP):

Service name: HTTP Apache

Version detected: 2.2.8

Server header: HTTP/1.1 200 OK

Potential vulnerabilities: XSS, weak default TLS settings

Next steps: run vulnerability scan, solve/upgrade known or found issues

Port 3306 (MySQL - if found):

Service name: n/a


```

Port 3306 (MySQL - if found):
Service name: n/a
Version detected:n/a
Security concerns:n/a

TARGET 2 (Windows - Metasploitable3):
=====
Port 445 (SMB):
Service name: microsoft-ds
Version detected: Microsoft Windows server 2008 R2
OS version hints: Windows NT 6.1
Known vulnerabilities: SMB signing disables (vulnerable to MITM attacks)

Port 3389 (RDP):
Service name: tcpwrapped
Version detected:n/a
Security notes:n/a
Encryption level:n/a

Port 135 (RPC - if scanned):
Service name:
Version detected:
Notes:

```

ANALYSIS QUESTIONS:

1. Which services appear to be running outdated versions?

ANALYSIS QUESTIONS:

```

1. Which services appear to be running outdated versions?
FTP, OpenSSH, HTTP, Microsoft-ds
2. Which port presents the highest security risk? Why?
Microsoft-ds (port 445) presents the highest security risk because
it's hardest to defend and harden. It's commonly open and misconfigured on internal networks
making it ideal for lateral movement, data theft, and ransomware. SMB also often runs with high
privileges and is integrated deeply with Windows.
3. What would you research next based on these findings?
The best way to harden and restrict Microsoft-ds on port 445, most likely being update/upgrade. Then I would address
other vulnerabilities on other ports.
4. How does Nmap's version detection compare to manual banner grabbing?
nmap is deeper and often more reliable but banner grabbing is useful because it's quick and produces less network noise.

```

Checkpoint Questions:

1. Because then you're able to look into vulnerabilities for Apache 2.2.8 and either attack or defend against them
2. -version-intensity 0 is a very light and fast scan while -version-intensity 9 is the maximum intensity and thorough scan
3. Manual banner grabbing might work sometimes when nmap -sV fails because it is a single, more simple interaction while nmap -sV uses lots of automated probes that could be potentially blocked or mishandled.

Exercise 4:

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
```

```
$ echo "=== Basic OS Detection ==="
```

```
sudo nmap -O $TARGET1_IP
```

```
=== Basic OS Detection ===
```

```
[sudo] password for quintin:
```

```
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-31 10:01 EDT
```

```
Nmap scan report for 192.168.1.100
```

```
Host is up (0.0012s latency).
```

```
Not shown: 977 closed tcp ports (reset)
```

PORT	STATE	SERVICE
21/tcp	open	ftp
22/tcp	open	ssh
23/tcp	open	telnet
25/tcp	open	smtp
53/tcp	open	domain
80/tcp	open	http
111/tcp	open	rpcbind
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
512/tcp	open	exec
513/tcp	open	login
514/tcp	open	shell
1099/tcp	open	rmiregistry
1524/tcp	open	ingreslock
2049/tcp	open	nfs
2121/tcp	open	ccproxy-ftp
3306/tcp	open	mysql
5432/tcp	open	postgresql
5900/tcp	open	vnc
6000/tcp	open	X11

```
8180/tcp open  unknown
```

```
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

```
Device type: general purpose
```

```
Running: Linux 2.6.X
```

```
OS CPE: cpe:/o:linux:linux_kernel:2.6
```

```
OS details: Linux 2.6.9 - 2.6.33
```

```
Network Distance: 1 hop
```

```
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

```
Nmap done: 1 IP address (1 host up) scanned in 1.97 seconds
```

```
└─$ echo "=== Aggressive OS Guessing ==="
sudo nmap -O --osscan-guess $TARGET1_IP
=== Aggressive OS Guessing ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-31 10:06 EDT
Nmap scan report for 192.168.1.100
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
```

MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Device type: general purpose

Running: Linux 2.6.X

OS CPE: cpe:/o:linux:linux_kernel:2.6

OS details: Linux 2.6.9 - 2.6.33

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 1.99 seconds

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ cat os_detection_linux.txt
# Nmap 7.95 scan initiated Fri Oct 31 10:08:36 2025 as: /usr/lib/nmap/nmap -O -oN os_detection_linux.txt 192.168.1.100
Nmap scan report for 192.168.1.100
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ echo ""
echo "=== Windows Target OS Detection ==="
sudo nmap -O $TARGET2_IP -oN os_detection_windows.txt

=== Windows Target OS Detection ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-31 11:50 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.62 seconds
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ echo ""
echo "=== Windows Target OS Detection ==="
sudo nmap -O $TARGET2_IP -oN os_detection_windows.txt

=== Windows Target OS Detection ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-31 11:53 EDT
Nmap scan report for 192.168.1.105
Host is up (0.00092s latency).
Not shown: 982 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
3389/tcp  open  ms-wbt-server
4848/tcp  open  appserv-http
5985/tcp  open  wsman
7676/tcp  open  imqbrokerd
```



```
8080/tcp open  http-proxy
8181/tcp open  intermapper
8383/tcp open  m2mservices
9200/tcp open  wap-wsp
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
MAC Address: 08:00:27:D7:CC:D8 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Server 2008 R2 SP1 or Windows 7 SP1, Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Win
dows 8.1
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.55 seconds
```

```
└─$ echo ""
echo "=== Scanning Both Targets Simultaneously ==="
sudo nmap -A $TARGET1_IP $TARGET2_IP -oN both_targets_comprehensive.txt

=== Scanning Both Targets Simultaneously ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-31 12:08 EDT
Nmap scan report for 192.168.1.100
Host is up (0.00096s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.1.103
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
| sslv2:
```

```
8180/tcp open  http          Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:52:28:6B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Host script results:

```
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2025-10-29T05:38:31-04:00
|_ smb2-time: Protocol negotiation failed (SMB2)
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_ clock-skew: mean: -2d05h31m09s, deviation: 2h00m03s, median: -2d06h31m11s
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
```

TRACEROUTE

HOP	RTT	ADDRESS
-----	-----	---------

Nmap scan report for 192.168.1.105

Host is up (0.00086s latency).

Not shown: 982 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

21/tcp	open	ftp	Microsoft ftpd
--------	------	-----	----------------

|_ ftp-syst:

|_ SYST: Windows_NT

80/tcp	open	http	Microsoft IIS httpd 7.5
--------	------	------	-------------------------

|_ http-methods:

|_ Potentially risky methods: TRACE

|_ http-server-header: Microsoft-IIS/7.5

|_ http-title: Site doesn't have a title (text/html).

135/tcp	open	msrpc	Microsoft Windows RPC
---------	------	-------	-----------------------

139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
---------	------	-------------	-------------------------------

445/tcp	open	microsoft-ds	Windows Server 2008 R2 Standard 7601 Service Pack 1 microsoft-ds
---------	------	--------------	--

3306/tcp	open	mysql	MySQL 5.5.20-log
----------	------	-------	------------------

|_ mysql-info:

|_ Protocol: 10

|_ Version: 5.5.20-log

|_ Thread ID: 6

|_ Capabilities flags: 63487

|_ Some Capabilities: Support41Auth, LongPassword, Speaks41ProtocolOld, SupportsTransactions, Speaks41ProtocolNew, IgnoreSpaceBeforeParenth

esis, DontAllowDatabaseTableColumn, LongColumnFlag, SupportsCompression, FoundRows, ConnectWithDatabase, IgnoreSigpipes, InteractiveClient,

ODBCClient, SupportsLoadDataLocal, SupportsMultipleResults, SupportsAuthPlugins, SupportsMultipleStatements

|_ Status: Autocommit

|_ Salt: v6E<;SnS3E@RYLN6lrVz

|_ Auth Plugin Name: mysql_native_password

3389/tcp	open	tcpwrapped	
----------	------	------------	--

|_ ssl-date: 2025-10-31T16:10:09+00:00; +6s from scanner time.

|_ ssl-cert: Subject: commonName=vagrant-2008R2

|_ Not valid before: 2025-08-24T17:10:11

| smb-security-mode:

|_ account_used: <blank>

|_ authentication_level: user

|_ challenge_response: supported

|_ message_signing: disabled (dangerous, but default)

|_ smb2-time:

|_ date: 2025-10-31T16:09:44

|_ start_date: 2025-10-31T15:52:32

|_ clock-skew: mean: 1h00m07s, deviation: 2h38m46s, median: 6s

|_ smb-os-discovery:

|_ OS: Windows Server 2008 R2 Standard 7601 Service Pack 1 (Windows Server 2008 R2 Standard 6.1)

|_ OS CPE: cpe:/o:microsoft:windows_server_2008::sp1

|_ Computer name: vagrant-2008R2

|_ NetBIOS computer name: VAGRANT-2008R2\x00

|_ Workgroup: WORKGROUP\x00

|_ System time: 2025-10-31T09:09:47-07:00

|_ nbstat: NetBIOS name: VAGRANT-2008R2, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:d7:cc:d8 (PCS Systemtechnik/Oracle VirtualBox virtual

NIC)

|_ smb2-security-mode:

|_ 2:1:0:

|_ Message signing enabled but not required

TRACEROUTE

HOP	RTT	ADDRESS
-----	-----	---------

1	0.86 ms	192.168.1.105
---	---------	---------------

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 2 IP addresses (2 hosts up) scanned in 94.95 seconds

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ echo "=== Analyzing Comprehensive Scan Results ==="
echo ""
echo "1. Service Versions Found:"
grep "open" comprehensive_linux.txt
echo ""
echo "2. OS Detection:"
grep -A 5 "OS details:" comprehensive_linux.txt
echo ""
echo "3. Traceroute Results:"
grep -A 10 "TRACEROUTE" comprehensive_linux.txt
echo ""
echo "4. NSE Script Results:"
grep "Script" comprehensive_linux.txt
echo "=== Analyzing Comprehensive Scan Results ==="
```

1. Service Versions Found:

21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
80/tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp	open	exec	netkit-rsh rexecd
513/tcp	open	login	
514/tcp	open	tcpwrapped	
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell

1524/tcp	open	bindshell	Metasploitable root shell
2049/tcp	open	nfs	2-4 (RPC #100003)
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	VNC (protocol 3.3)
6000/tcp	open	X11	(access denied)
6667/tcp	open	irc	UnrealIRCd
8009/tcp	open	ajp13	Apache Jserv (Protocol v1.3)
8180/tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1

2. OS Detection:

OS details: Linux 2.6.9 - 2.6.33

Network Distance: 1 hop

Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:

| smb-os-discovery:

3. Traceroute Results:

TRACEROUTE

HOP	RTT	ADDRESS
1	1.17 ms	192.168.1.100

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done at Fri Oct 31 11:48:02 2025 -- 1 IP address (1 host up) scanned in 24.38 seconds

4. NSE Script Results:

```

(quintin@kali)-[~/cybersec-labs/nmap-lab]
└─$ echo "=== OS Detection Comparison ==="
echo ""
echo "Target 1 (Linux):"
grep "OS details:" os_detection_linux.txt
echo ""
echo "Target 2 (Windows):"
grep "OS details:" os_detection_windows.txt
echo ""
echo "Can you explain how you could tell them apart just from their OS detection
results?"
=== OS Detection Comparison ===

Target 1 (Linux):
OS details: Linux 2.6.9 - 2.6.33

Target 2 (Windows):
OS details: Microsoft Windows Server 2008 R2 SP1 or Windows 7 SP1, Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Win
dows 8.1

Can you explain how you could tell them apart just from their OS detection
results?

```

Checkpoint Questions:

1. OS detection requires root privileges because it needs to send and receive raw packets, which is a privileged operation that normal users aren't allowed to do
2. The -O is just used for OS detection while the -A flag combines multiple scan types for maximum information like OS detection, version detection, script scanning, and traceroute.
3. The target is on the same local network or directly adjacent, no routers between.

Exercise 5:


```
(quintin@kali)-[~]  
$ echo "Total NSE scripts installed:"  
ls /usr/share/nmap/scripts/ | wc -l  
Total NSE scripts installed:  
610
```

```
(quintin@kali)-[~]  
$ ls /usr/share/nmap/scripts/ | head -20  
echo "... and many more"  
acarsd-info.nse  
address-info.nse  
afp-brute.nse  
afp-ls.nse  
afp-path-vuln.nse  
afp-serverinfo.nse  
afp-showmount.nse  
ajp-auth.nse  
ajp-brute.nse  
ajp-headers.nse  
ajp-methods.nse  
ajp-request.nse  
allseeingeve-info.nse  
amqp-info.nse  
asn-query.nse  
auth-owners.nse  
auth-spoof.nse  
backorifice-brute.nse  
backorifice-info.nse  
bacnet-info.nse  
... and many more
```

```
(quintin@kali)-[~]  
$ echo ""  
echo "=== FTP Scripts ==="  
ls /usr/share/nmap/scripts/ | grep -i ftp
```

```
=== FTP Scripts ===  
ftp-anon.nse  
ftp-bounce.nse  
ftp-brute.nse  
ftp-libopie.nse  
ftp-proftpd-backdoor.nse  
ftp-syst.nse  
ftp-vsftpd-backdoor.nse  
ftp-vuln-cve2010-4221.nse  
tftp-enum.nse  
tftp-version.nse
```

```
(quintin@kali)-[~]  
$ echo ""  
echo "=== SMB Scripts ==="  
ls /usr/share/nmap/scripts/ | grep -i smb
```

```
=== SMB Scripts ===  
smb2-capabilities.nse  
smb2-security-mode.nse  
smb2-time.nse  
smb2-vuln-uptime.nse  
smb-brute.nse  
smb-double-pulsar-backdoor.nse  
smb-enum-domains.nse  
smb-enum-groups.nse
```

```
echo "=== HTTP Scripts ==="
ls /usr/share/nmap/scripts/ | grep -i http | head -10
echo "... and more"
```

```
=== HTTP Scripts ===
http-adobe-coldfusion-apsa1301.nse
http-affiliate-id.nse
http-apache-negotiation.nse
http-apache-server-status.nse
http-aspnet-debug.nse
http-auth-finder.nse
http-auth.nse
http-avaya-ipoffice-users.nse
http-awstatstotals-exec.nse
http-axis2-dir-traversal.nse
... and more
```

```
(quintin@kali)-[~]
$ echo "=== FTP Anonymous Login Check ==="
nmap --script-help ftp-anon
echo ""
=== FTP Anonymous Login Check ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 19:21 EDT
```

```
ftp-anon
Categories: default auth safe
https://nmap.org/nsedoc/scripts/ftp-anon.html
Checks if an FTP server allows anonymous logins.
```

If anonymous is allowed, gets a directory listing of the root directory and highlights writeable files.

```
(quintin@kali)-[~]
$ echo "=== SMB OS Discovery ==="
nmap --script-help smb-os-discovery
echo ""
# Learn about HTTP enumeration
echo "=== HTTP Directory Enumeration ==="
nmap --script-help http-enum
=== SMB OS Discovery ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 19:21 EDT

smb-os-discovery
Categories: default discovery safe
https://nmap.org/nsedoc/scripts/smb-os-discovery.html
Attempts to determine the operating system, computer name, domain, workgroup, and current
time over the SMB protocol (ports 445 or 139).
This is done by starting a session with the anonymous
account (or with a proper user account, if one is given; it likely doesn't make
a difference); in response to a session starting, the server will send back all this
information.

The following fields may be included in the output, depending on the
circumstances (e.g. the workgroup name is mutually exclusive with domain and forest
names) and the information available:
* OS
* Computer name
* Domain name
* Forest name
* FQDN
* NetBIOS computer name
* NetBIOS domain name
* Workgroup
```

```
(quintin@kali)-[~]
$ cat nse_default_scan.txt
# Nmap 7.95 scan initiated Sat Nov 1 19:43:43 2025 as: /usr/lib/nmap/nmap --privileged -sC -oN nse_default_scan.txt 192.168.1.100
Nmap scan report for 192.168.1.100
Host is up (0.00098s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 192.168.1.103
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   vsFTPD 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet
25/tcp    open  smtp
| sslv2:
|   SSLv2 supported
| ciphers:
```



```

(quintin@kali)-[~]
$ grep ".* nse_default_scan.txt
ftp-syst:
STAT:
FTP server status:
  Connected to 192.168.1.103
  Logged in as ftp
  TYPE: ASCII
  No session bandwidth limit
  Session timeout in seconds is 300
  Control connection is plain text
  Data connections will be plain text
  vsFTPD 2.3.4 - secure, fast, stable
_End of status
ftp-anon: Anonymous FTP login allowed (FTP code 230)
ssh-hostkey:
  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
sslv2:
  SSLv2 supported
  ciphers:
    SSL2_DES_64_CBC_WITH_MD5
    SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
    SSL2_RC4_128_EXPORT40_WITH_MD5
    SSL2_RC2_128_CBC_WITH_MD5
    SSL2_RC4_128_WITH_MD5
    SSL2_DES_192_EDE3_CBC_WITH_MD5
ssl-date: 2025-11-01T23:46:36+00:00; +1m46s from scanner time.
ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/cou
ntryName=XX

```

```

(quintin@kali)-[~]
$ echo "=== FTP Enumeration Scripts ==="
# Check for anonymous FTP, FTP bounce attack, and vsftpd backdoor
nmap --script ftp-anon,ftp-bounce,ftp-vsftpd-backdoor -p 21 $TARGET1_IP
# Save results
nmap --script ftp-anon,ftp-bounce,ftp-vsftpd-backdoor -p 21 $TARGET1_IP -oN ftp_scripts.txt
=== FTP Enumeration Scripts ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 20:02 EDT
NSE: [ftp-bounce] Couldn't resolve scanme.nmap.org, scanning 10.0.0.1 instead.
NSE: [ftp-bounce] PORT response: 500 Illegal PORT command.
Nmap scan report for 192.168.1.100
Host is up (0.0025s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDs:  CVE:CVE-2011-2523  BID:48539
|         vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root)
|       References:
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|         https://www.securityfocus.com/bid/48539
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)

```

```

└─$ echo ""
echo "=== SMB Enumeration on Windows Target ==="
# Discover SMB OS, check security mode
nmap --script smb-os-discovery,smb-security-mode -p 445 $TARGET2_IP -oN smb_scripts.txt
echo ""
echo "=== HTTP Enumeration ==="
# Enumerate web directories, check HTTP headers and methods
nmap --script http-enum,http-headers,http-methods,http-robots.txt \
-p 80 $TARGET1_IP -oN http_scripts.txt
echo ""
echo "=== SSH Enumeration ==="
# Get host key and supported authentication methods
nmap --script ssh-hostkey,ssh-auth-methods -p 22 $TARGET1_IP -oN ssh_scripts.txt

=== SMB Enumeration on Windows Target ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 20:09 EDT
Nmap scan report for 192.168.1.106
Host is up (0.0011s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:D7:CC:D8 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb-os-discovery:
|   OS: Windows Server 2008 R2 Standard 7601 Service Pack 1 (Windows Server 2008 R2 Standard 6.1)

```

```

└─$ echo "=== All FTP Scripts ==="
nmap --script "ftp-*" -p 21 $TARGET1_IP
=== All FTP Scripts ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 20:11 EDT
NSE: [ftp-bounce] Couldn't resolve scanme.nmap.org, scanning 10.0.0.1 instead.
NSE: [ftp-bounce] PORT response: 500 Illegal PORT command.
NSE: [ftp-brute] usernames: Time limit 10m00s exceeded.
NSE: [ftp-brute] usernames: Time limit 10m00s exceeded.
NSE: [ftp-brute] passwords: Time limit 10m00s exceeded.
Nmap scan report for 192.168.1.100
Host is up (0.0011s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPD version 2.3.4 backdoor
|     State: VULNERABLE (Exploitable)
|     IDs: BID:48539 CVE:CVE-2011-2523
|     vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|     Disclosure date: 2011-07-03
|     Exploit results:
|       Shell command: id
|       Results: uid=0(root) gid=0(root)
|     References:
|       https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|       https://www.securityfocus.com/bid/48539
|       http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|_ ftp-brute:

```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ echo "=== All HTTP Scripts ==="
nmap --script "http-*" -p 80 $TARGET1_IP
=== All HTTP Scripts ===
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-02 20:51 EST
Pre-scan script results:
|_http-robtex-shared-ns: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
Nmap scan report for 192.168.1.103
Host is up (0.00011s latency).

PORT      STATE SERVICE
80/tcp    closed http

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

```
(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ ./nse_comprehensive.sh

=====
Comprehensive NSE Scanning
=====

This script will:
1. Enumerate services on both targets
2. Run targeted NSE scripts per service
3. Check for known vulnerabilities
4. Save all results to separate files

Starting time: Sun Nov  2 09:11:57 PM EST 2025

=====
TARGET 1: Linux/Metasploitable
=====

— Scanning FTP (Port 21) —
Checking for: anonymous login, backdoors, bounce attacks
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-02 21:11 EST
Nmap scan report for 192.168.1.103
Host is up (0.000040s latency).

PORT      STATE SERVICE
21/tcp    closed ftp

Nmap done: 1 IP address (1 host up) scanned in 0.31 seconds
✓ Results saved to nse_results/ftp.txt
```

Generating Summary Report

NSE Scan Summary Report

Scan Date: Sun Nov 2 09:16:31 PM EST 2025

Targets Scanned:

Target 1 (Linux): 192.168.1.103
Target 2 (Windows): 192.168.1.106

CRITICAL FINDINGS:

FTP Anonymous Login:
No results

FTP Backdoor (vsftpd 2.3.4):

Nmap 7.95 scan initiated Sun Nov 2 21:11:57 2025 as: /usr/lib/nmap/nmap --privileged --script ftp-anon,ftp-bounce,ftp-vsftpd-backdoor,ftp-proftpd-backdoor -p 21 -oN nse_results/ftp.txt 192.168.1.103
Nmap scan report for 192.168.1.103
Host is up (0.000040s latency).

SMB Vulnerabilities:

| VULNERABLE:
| State: VULNERABLE

HTTP Vulnerabilities:

MySQL Databases:

No
results

DETAILED RESULTS:

All detailed results saved in nse_results/ directory:
total 44K

-rw-rw-r--	1	quintin	quintin	403	Nov	2	21:11	ftp.txt
-rw-rw-r--	1	quintin	quintin	482	Nov	2	21:12	mssql.txt
-rw-rw-r--	1	quintin	quintin	504	Nov	2	21:11	postgresql.txt
-rw-rw-r--	1	quintin	quintin	472	Nov	2	21:12	rdp.txt
-rw-rw-r--	1	quintin	quintin	1.2K	Nov	2	21:12	smb_vuln.txt
-rw-rw-r--	1	quintin	quintin	387	Nov	2	21:11	ssh.txt
-rw-rw-r--	1	quintin	quintin	1.5K	Nov	2	21:16	SUMMARY_REPORT.txt
-rw-rw-r--	1	quintin	quintin	383	Nov	2	21:11	telnet.txt
-rw-rw-r--	1	quintin	quintin	419	Nov	2	21:12	vuln_linux.txt
-rw-rw-r--	1	quintin	quintin	4.5K	Nov	2	21:16	vuln_windows.txt

Scanning Complete

Completion time: Sun Nov 2 09:16:31 PM EST 2025

All results saved to nse_results/ directory
Summary report: nse_results/SUMMARY_REPORT.txt

NSE Scan Analysis

Analyst: Quintin Applegate

Date: 11/2/25

HIGH-RISK FINDINGSTarget 1 (Linux - Metasploitable):☐ FTP Anonymous Login:

Status: Disabled

Impact: N/A

Evidence:

PORT	STATE	SERVICE
21/tcp	closed	ftp

Recommendation: Keep port closed

☐ vsftpd 2.3.4 Backdoor:

Status: Not vulnerable

Impact: N/A

Evidence:

PORT	STATE	SERVICE
21/tcp	closed	ftp

CVE: CVE-2011-2523

Recommendation: Keeping port closed will continue to eliminate vulnerability

☐ Telnet Service:

□ Telnet Service:

Status: Not running

Impact: N/A [Unencrypted remote access - credentials sent in cleartext]

Evidence:

PORT STATE SERVICE

23/tcp closed telnet

Recommendation: Keep port closed, vulnerability not present

□ HTTP Vulnerabilities:

Shellshock: Not Vulnerable

SQL Injection: Not Vulnerable

XSS: Not Vulnerable

Evidence:

PORT STATE SERVICE

80/tcp closed http

Recommendation: Keeping port 80 closed will keep vulnerability out of scope

□ MySQL Empty Password:

Status: [Root has empty password: No]

Impact: N/A [Unauthenticated database access]

Evidence:

PORT STATE SERVICE

3306/tcp closed mysql

Recommendation: Keep port closed, then no SQL vulnerability

Target 2 (Windows - Metasploitable3):

Target 2 (Windows - Metasploitable3):

□ SMB Signing Disabled:

Status: Enabled

Impact: [Vulnerable to man-in-the-middle attacks]

Evidence: message_signing: disabled (dangerous, but default)

Recommendation: Turn SMB Message Signing Disabled from Disabled to Enabled

□ EternalBlue (MS17-010):

Status: Vulnerable

Impact: [Remote code execution with SYSTEM privileges]

Evidence: Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)

| State: VULNERABLE

| IDs: CVE:CVE-2017-0143

| Risk factor: HIGH

| A critical remote code execution vulnerability exists in Microsoft SMBv1

| servers (ms17-010)

CVE: CVE-2017-0143

Recommendation: Implement security update that addresses the vulnerability by correcting how SMBv1 handles these specially crafted requests

□ MS08-067 Vulnerability:

Status: Patched

Impact: [Remote code execution]

Evidence:

Host script results:

|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED

MEDIUM-RISK FINDINGS

Target 1:

□ Exposed SMB Shares:

Shares found:

ERROR: Enumerating shares failed, guessing at common ones (NT_STATUS_ACCESS_DENIED)

```
\\192.168.1.106\ADMIN$:
|   warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|   Anonymous access: <none>
| \\192.168.1.106\C$:
|   warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|   Anonymous access: <none>
| \\192.168.1.106\IPC$:
|   warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|_  Anonymous access: READ
```

Permissions: Read and none

Impact: None-Low impact, shows potential threat actor access could be granted with correct credentials if found, however currently denies access. Having port 445 open and SMB open to untrusted networks presents a small risk as well.

□ HTTP Directory Enumeration:

Directories found: No results found

Sensitive paths exposed: No results found

□ HTTP Directory Enumeration:
Directories found: No results found
Sensitive paths exposed: No results found
Impact: N/A

□ Database Information Disclosure:
Databases found: N/A port closed
Version info exposed: N/A
Impact: N/A

Target 2:

□ SMB User Enumeration:
Users found: None
Impact: [Username list for brute force attacks]

□ MS-SQL Configuration Issues:
Issues found: N/A port closed
Impact: N/A port closed

INFORMATIONAL FINDINGS

Target 1:

□ SSH Version: Unfound, closed port
□ HTTP Server: Unfound, closed port

Target 1:

- SSH Version: Unfound, closed port
- HTTP Server: Unfound, closed port
- MySQL Version: Unfound, closed port
- Supported SSH Auth Methods: Unfound, closed port

Target 2:

- Windows Version: Windows Server 2008 R2 Standard 7601 Service Pack 1 (Windows Server 2008 R2 Standard 6.1)
- SMB Version: Unable to find from smb_enum.txt, however OSINT allowed me to find out either SMB 2.0/2.1 since those are the only 2 that were used for Windows Server 2008 R2-- the OS found from the scan.
- Domain/Workgroup: WORKGROUP\x00
- Computer Name: VAGRANT-2008R2\x00

ATTACK PATH ANALYSIS

Based on NSE findings, identify potential attack vectors:

Target 1 Attack Path:

1. Exploit weak RDP encryption that is using TLS 1.0 by using BEAST or POODLE attacks

Target 2 Attack Path:

1. Exploit EternalBlue (MS17-010) for initial access
2. Exploit SMB signing being disabled with a MITM attack

REMEDIATION PRIORITIES

CRITICAL (Fix Immediately):

1. Require SMB signing
2. Implement security update that addresses the vulnerability by correcting how SMBv1 handles these specially crafted requests
3. Upgrade to TLS 1.2 and restrict/tunnel RDP through a VPN or gateway over either HTTPS or TLS 1.2. Ideally, the OS would be upgraded to a newer version, but these recommended actions are assuming that the host will stay on the current OS.

HIGH (Fix Within 1 Week):

1. Block inbound traffic on port 445 at the network perimeter right now
2. Enable Kerberos authentication for port 445
- 3.

MEDIUM (Fix Within 1 Month):

- 1.
- 2.
- 3.

NEXT STEPS

Based on this NSE scan, recommended next steps:

1. Exploitation Phase:
 - Which vulnerability to exploit first:
 - Tool to use (Metasploit/manual):
 - Expected outcome:
2. Further Enumeration:

```

(quintin@kali)-[~/cybersec-labs/nmap-lab]
$ ./nse_challenge.txt
=== NSE Challenge ===

Using only NSE scripts, answer these questions:

1. Does the Linux target allow anonymous FTP login?
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-03 23:49 EST
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 192.168.1.103
Host is up (0.000048s latency).

PORT      STATE SERVICE
21/tcp    closed ftp

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds
Answer: No

2. Is the Windows target vulnerable to EternalBlue (MS17-010)?
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-03 23:49 EST
Nmap scan report for 192.168.1.106
Host is up (0.0014s latency).

PORT      STATE SERVICE
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:D7:CC:D8 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| smb-vuln-ms17-010:
|   VULNERABLE:
|
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|   State: VULNERABLE
|   IDs: CVE:CVE-2017-0143
|   Risk factor: HIGH
|   A critical remote code execution vulnerability exists in Microsoft SMBv1
|   servers (ms17-010).
|
|   Disclosure date: 2017-03-14
|   References:
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|   https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|   https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|_

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds
Answer: Yes

3. What HTTP directories are exposed on the Linux target?
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-03 23:49 EST
Nmap scan report for 192.168.1.103
Host is up (0.000039s latency).

PORT      STATE SERVICE
80/tcp    closed http

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
Answer: None

Run these commands and document your findings

```

Checkpoint Questions:

1. Default scripts are safe, useful, common enumeration scripts that won't hurt the target. Safe scripts are useful for staying stealthy since they are unlikely to trigger any alarms. Vuln scripts check for specific vulnerabilities
2. You would use `--script=ftp-anon` when you want fast, reliable, repeatable, and automated checks for anonymous FTP. It's highly scalable instead of manually checking for anonymous FTP on individual targets. However, manual checking for anonymous FTP is stealthier because it sends a single (lower noise) TCP connect sequence.
3. When an NSE script says "VULNERABLE" that means the script found a vulnerability for a system that could potentially be exploited.
4. You can use `nmap --script-help <script-name>` and look at the description, categories, usages, and references.

Knowledge Assessment:

1. B
2. True
3. A
4. D
5. C
6. D
7. False
8. B
9. B
10. B

Conclusion: After completing this lab, I can reliably discover live hosts, enumerate open ports and service versions, and identify attack vectors using Nmap and NSE. I can successfully perform OS fingerprinting, tune scan timing and intensity for speed or stealth, and automate checks for common vulnerabilities, then document my findings in professional, real-world like reports. These enumeration skills form the foundation for effective security assessments—providing the actionable intelligence I need for remediation, follow-up testing, and down the road, effective communication with stakeholders.